

Supplemental material

Table S1. Alcohol-related causes, source of relative risk, and ICD-10 codes

Causes	Source	ICD-10 Codes
Neuro-psychiatric		
Mental and behavioral disorders due to alcohol	AAF=1	F10
Degeneration of nervous system due to alcohol	AAF=1	G312
Epilepsy	Samokhvalov (2010) ¹	G40-G41
Alcohol polyneuropathy	AAF=1	G621
Cardiovascular		
Hypertensive disease	WHO (2018) ²	I10-I15
Ischemic heart disease	Rehm (2016) ³	I20-I25
Alcoholic cardiomyopathy	AAF=1	I426
Cardiac arrhythmias	AAF=1	I48X
Hemorrhagic and other non-ischemic stroke	Larsson (2016) ⁴	I60-I62
Ischemic stroke	Rehm (2016) ³	I63-I66
Cancer		
Mouth, oropharynx and laryngeal cancer	Bagnardi (2013) ⁵	C00-C14
Esophageal cancer	Bagnardi (2013) ⁵	C15
Colon cancer	Bagnardi (2013) ⁵	C18
Colorectal cancer	Bagnardi (2013) ⁵	C19X
Rectal cancer	Bagnardi (2013) ⁵	C20
Liver cancer	Turati (2014) ⁶	C22
Laryngeal cancer	Bagnardi (2013) ⁵	C32
Breast cancer	Bagnardi (2013) ⁵	C50
Other causes attributable to alcohol		
Non-insulin-dependent diabetes mellitus	Knott (2015) ⁷	E11
Tuberculosis	Imtiaz (2017) ⁸	A15-A19
HIV/AIDS	Rehm (2017) ⁹	B20-B24
Pneumonia	Samokhvalov (2010) ¹	J12-J18
Alcoholic gastritis	AAF=1	K292
Cirrhosis of the liver	WHO (2018) ²	K70, K74
Alcohol-induced chronic pancreatitis	AAF=1	K860
Acute and chronic pancreatitis	Samokhvalov (2015) ¹⁰	K85, K861
Injuries		
Intentional self-poisoning by exposure to alcohol	AAF = 1	X65
Accidental poisoning by exposure to alcohol	AAF = 1	X45
Poisoning by exposure to alcohol, undermined intent	AAF = 1	Y15
Motor vehicle accidents	WHO (2018) ²	a
Accidental poisoning	WHO (2018) ²	X40, X46-X49
Falls	WHO (2018) ²	W00-W19
Fires	WHO (2018) ²	X00-X09
Drowning	WHO (2018) ²	W65-W74
Other Unintentional injuries	WHO (2018) ²	b
Self-inflicted injuries	WHO (2018) ²	X60-X64, X66-X84,

Homicide	WHO (2018) ²	Y87.0
Other intentional injuries	WHO (2018) ²	X85-Y09, Y87.1
		Y35

AAF = Alcohol-Attributable Fraction. WHO = World Health Organization

a V021–V029, V031–V039, V041–V049, V092, V093, V123–V129, V133–V139, V143–V149, V194–V196, V203–V209, V213–V219, V223–V229, V233–V239, V243–V249, V253–V259, V263–V269, V273–V279, V283–V289, V294–V299, V304–V309, V314–V319, V324–V329, V334–V339, V344–V349, V354–V359, V364–V369, V374–V379, V384–V389, V394–V399, V404–V409, V414–V419, V424–V429, V434–V439, V444–V449, V454–V459, V464–V469, V474–V479, V484–V489, V494–V499, V504–V509, V514–V519, V524–V529, V534–V539, V544–V549, V554–V559, V564–V569, V574–V579, V584–V589, V594–V599, V604–V609, V614–V619, V624–V629, V634–V639, V644–V649, V654–V659, V664–V669, V674–V679, V684–V689, V694–V699, V704–V709, V714–V719, V724–V729, V734–V739, V744–V749, V754–V759, V764–V769, V774–V779, V784–V789, V794–V799, V803–V805, V811, V821, V830–V833, V840–V843, V850–V853, V860–V863, V870–V878, V892. **b** W20–W64, W75–W99, X10–X39, X50–X59, Y40–Y86, Y88, Y89 and V series except for the codes listed in note a.

chronic pancreatitis												
Acute pancreatitis	19	17;20	19	18;21	12	11;13	13	11;15	16	15;18	15	14;16
Injuries												
Intentional self poisoning	0	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1
Accidental poisoning	3	2;4	3	2;4	1	1;2	3	2;4	2	1;3	5	4;6
Motor vehicle accident	30	16;45	21	11;33	19	10;30	26	14;40	29	16;46	67	40;93
Falls	3	2;4	4	3;5	4	3;5	7	6;8	5	4;6	11	9;12
Fires	11	9;13	7	6;9	5	4;6	13	10;15	6	5;7	12	10;14
Drowning	8	7;10	5	4;6	2	1;3	6	5;7	3	2;4	9	8;10
Other unintentional injuries	20	17;24	42	34;51	10	8;12	35	29;41	21	17;25	57	49;64
Homicide	18	15;21	19	15;23	14	11;17	23	19;27	15	13;18	20	17;22

Table S2 shows that liver cirrhosis consistently contributed the greatest number of alcohol-attributable deaths among women, peaking at 768 in 2008 and decreasing to 516 in 2022, with some fluctuations over time. Injuries, particularly motor vehicles and other unintentional injuries were significant contributors among younger women. Motor vehicle accidents increased in 2018, with 120 deaths, but declined to 26 in 2022. Other unintentional injuries also peaked in 2018 before declining in 2022. Among cardiovascular causes, ischemic heart disease was the leading contributor, showing an increasing trend. Compared with other categories, neuropsychiatric causes contribute fewer deaths, with mental and behavioral disorders due to alcohol showing a slight increase toward 2022.

Table S3. Alcohol-Attributable Mortality for males 15-to 65 years old between 2008 to 2018

Causes	2008		2010		2012		2014		2016		2018	
	n	95%CI	n	95%CI	n	95%CI	n	95%CI	n	95%CI	n	95%CI
Neuro-Psychiatric												
Mental and Behavioral disorders	161	AAF = 1	118	AAF = 1	115	AAF = 1	67	AAF = 1	72	AAF = 1	77	AAF = 1
Degeneration of nervous system	14	AAF = 1	23	AAF = 1	13	AAF = 1	7	AAF = 1	6	AAF = 1	6	AAF = 1
Epilepsy	9	7;12	8	6;11	7	5;9	11	8; 15	7	5;9	11	8;14
Cardiovascular												
Hypertensive Disease	60	38;84	48	30;70	50	31;72	75	48;105	57	36;81	52	33;74
Ischemic Heart Disease	1192	968;1397	1158	940;1356	1175	954;1376	1324	1075;1551	1209	982;1416	1240	1007;1453
Cardiomyopathy	9	AAF = 1	4	AAF = 1	7	AAF = 1	5	AAF = 1	8	AAF = 1	7	AAF = 1
Hemorrhagic stroke	427	218;680	363	189;581	333	170;541	412	193;677	325	179;500	304	176;457
Ischemic stroke	24	16;32	24	16;32	19	13;26	21	15;29	23	16;31	20	13;26
Cancer												
Mouth, oropharynx	27	24;30	25	21;28	25	22;28	24	21;26	26	22;29	32	28;36
Esophageal	36	29;43	25	20;30	27	22;33	25	20;29	25	20;30	26	21;31
Colon, Rectal and Colorectal	94	89;98	87	82;91	86	81;91	113	106;119	137	131;143	157	150;163
Liver	62	59;65	57	53;60	73	68;77	59	56;63	74	70;77	75	71;78
Laryngeal	15	12;18	10	8;12	13	10;16	12	9;14	8	6;10	11	9;13
Other causes												
Diabetes Mellitus	15	11;24	14	12;21	19	14;29	17	13;25	17	15;23	15	14;18
Tuberculosis	48	9;85	43	8;80	47	8;87	36	7;66	31	5;59	33	6;62
HIV/AIDS	18	11;26	17	10;25	17	10;25	21	12;30	18	11;27	10	6;14
Pneumonia	28	6;51	30	6;57	23	5;44	30	6;57	25	5;47	29	6;54
Alcoholic gastritis	2	AAF = 1	3	AAF = 1	0	AAF = 1	0	AAF = 1	3	AAF = 1	3	AAF = 1

Liver Cirrhosis	1361	1262;1459	1285	1186;1383	1195	1097;1293	1284	1185;1385	1289	1195;1386	1063	988;1143
Alcohol-induced chronic pancreatitis	1	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1
Acute pancreatitis	46	35;58	39	31;50	38	29;49	50	37;63	39	31;49	40	32;50
Injuries												
Intentional self poisoning	0	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1	0	AAF = 1
Accidental poisoning	22	20;24	11	10;12	9	8;10	10	9;11	14	13;15	20	19;22
Motor vehicle accident	126	93;163	95	69;125	89	64;117	105	78;136	104	77;134	100	74;128
Falls	98	89;105	82	73;89	91	81;100	81	73;88	75	68;83	78	72;83
Fires	65	59;70	93	84;102	51	46;56	52	47;57	45	41;50	48	45;51
Drowning	164	150;175	112	101;123	97	87;107	96	87;104	98	88;106	109	101;116
Other unintentional injuries	394	361;423	382	343;418	281	252;309	413	373;448	295	265;323	544	503;578
Homicide	308	284;329	280	253;304	243	219;265	315	286;341	265	240;287	200	186;213

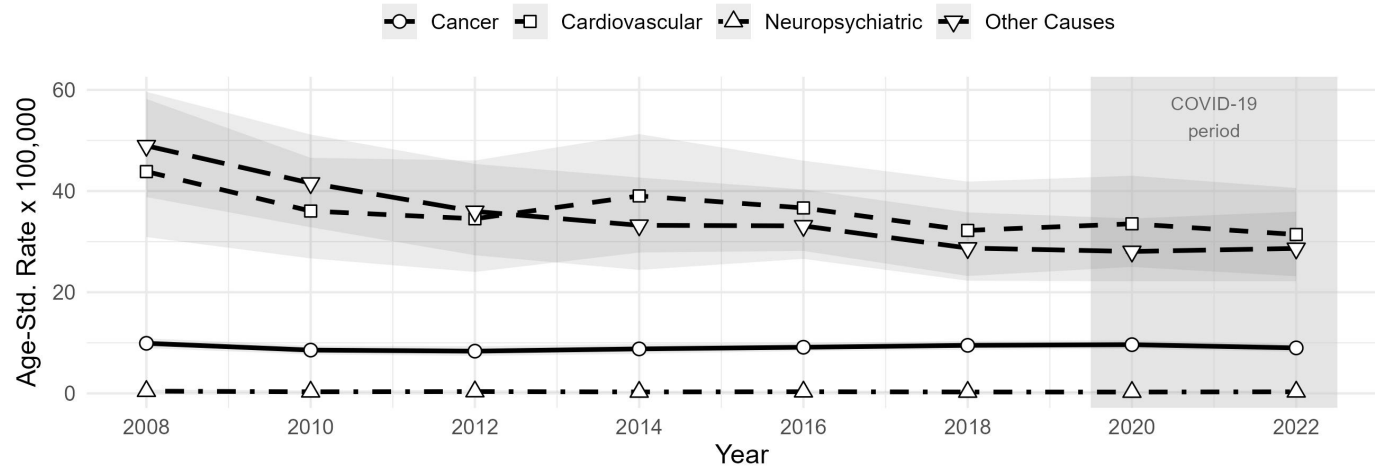
Table S3 displays alcohol-attributable mortality trends among men aged 15-65 years, revealing considerably higher mortality rates than in females across nearly all causes. Liver cirrhosis was consistently the leading contributor to alcohol-related deaths, but with fluctuating numbers, from 3,590 in 2008 to 2,938 in 2022. Injuries were prominent, with road injuries declining notably, whereas unintentional injuries peaked in 2018, with 1,146 alcohol-attributable deaths, and then decreased to 708 by 2022. Intentional injuries also exhibited a steady decline. Neuropsychiatric causes, particularly mental and behavioral disorders due to alcohol, fluctuated, with 263 deaths in 2008 and 256 in 2022. The prevalence of alcohol-attributable cancers, such as colon and rectum cancer and liver cancer, has shown an increasing trend, increasing from 210 in 2008 to 418 in 2022 for colorectal cancers and from 178 to 272 for liver cancers. Among cardiovascular causes, ischemic heart disease consistently accounted for the greatest number of deaths, increasing from 1,192 in 2008 to 1,470 in 2022.

Figure S1. Trends in alcohol-attributable mortality rates by injury subtype and non-injury causes, Chile 2008–2022

A) Injury subtypes



B) Non-injury causes



The line plot presented in Figure S1 shows age-standardized alcohol-attributable mortality rates by cause subtype from 2008 to 2022. Panel A shows injury subtypes, including road injuries, unintentional injuries, and intentional injuries. Road injuries declined markedly during the COVID-19 period (2020–2022), consistent with reduced mobility, while intentional injuries remained relatively stable over time. Unintentional injuries exhibited greater variability, including a notable peak in 2018.

Panel B presents non-injury causes, including cancer, cardiovascular diseases, neuropsychiatric conditions, and other causes. These categories show a general downward trend over the study period, with no clear deviation during the COVID-19 years relative to pre-existing trends.

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